This page is to be omitted in your proposal!

Note: This template is only an example. It shows what needs to be included for a typical project that includes implementation and system building. Different projects may require different format. Please consult your supervisor for the most suitable format.

The Project Title

Design and Specification Proposal
COMP702 — M.Sc. project (2024/25)

Submitted by
Your Name
(Your student number)

under the supervision of
Supervisor's Name

School of Computer Science and Informatics
University of Liverpool

1 Statement of ethical compliance: B2

Data Category: B

Participant Category: 2

I confirm that I have read the ethical guidelines and will follow them during this project. I will use anonymous data about humans from open source `https://www.kaggle.com/` in CA1 and human participants for requirement analysis and evaluation in CA3:

- CA1: Data Sources.
- CA3: Requirements Analysis; Evaluation.

2 Project description

In this project I plan to develop a system that can translate chip designs written in plain English into its formal specifications. The system will be efficient and accurate.

3 Aim and Requirements

3.1 The main aim

I will use Google's API to build a system that takes in plain English description of a chip design in voice and produces its formal specification in Verilog format.

3.2 Requirements

The requirements are:

- The system must be fast enough.
- The English must be plain enough.
- The system must be accurate enough.
- The verilog specification must be generated correctly.
- The system must be user-friendly.

If time permits, we will also implement some desirable feature such as:

- Verification of the generated Verilog specification.
- The system can also take in crude hand-drawn chip design and produce the Verilog specification.

4 Key literature and background reading

We will use the latest chip technology from [1] and combine it with the AI technique in [2]. There are two main components.

4.1 The medical component

The latest technique developed by Google in [1] ensures the correct translation from voice to text. We analyse it and discover that ....

4.2 The AI component

The latest AI technique involves technique AxZ that can be broken down as follows.

5 Preliminary work

In this section we will present the preliminary work that we have done so far. We have implemented the converter from sound to text using the latest Google API. We also have successfully trained the AI tool to translate the text into Verilog format. Preliminary testing shows that the system is very accurate and fast.

6 Development and Implementation Summary

Our work plan can be summarised as follows.

- Step 1: ...
- Step 2: ....

The interconnection between all these steps is illustrated in the following figure.

We will use some of the latest gadgets in our implementation such as:

- OpenAI.
- Azure.

7 Data source

We will use some dummy data.

8 Testing and evaluation

Describe how to test and evaluate the system.

9 UI/UX mockup

Present your initial design of UI/UX mockup here. Hand drawn figure is acceptable at this stage.

10 Project ethics and human participants

This project uses anonymised data humans from open source `https://www.kaggle.com/` in the first assessment. We will conduct survey where the participants are selected randomly to use our tool. Our use of data complies with the ethical guidance.

11 BCS project criteria

1. Application of Practical and Analytical Skills.

   This project will utilise practical and analytical skills from my degree, ...

2. Innovation and Creativity

   The project involves utilising some of the latest technology and combine them creatively....

12 Project plan

What is your plan to carry out the project? Give a Ghantt chart, if you like.

13 Risks and contingency plans

My risk management plan can be found in Table 1.

Risk	**Contingency**	**Likelihood**	**Impact**
Data is not clean	Clean up the data	High	High
Data does not fit	Find another data source	Medium	Low
Google conversion tool does not work	pay to use the premium version	low	high

Table 1: Risk management plan

References

[1] A.N. Author. The best paper of all. *Some really good reference* **12**, pp. 100–113, 2003.

[2] S. omeone. A paper that I wish I'd written. *J. Combinatorics* **34**, pp. 25–78, 1979.